HB-Adaptive model: learning diagnostics and data comparison

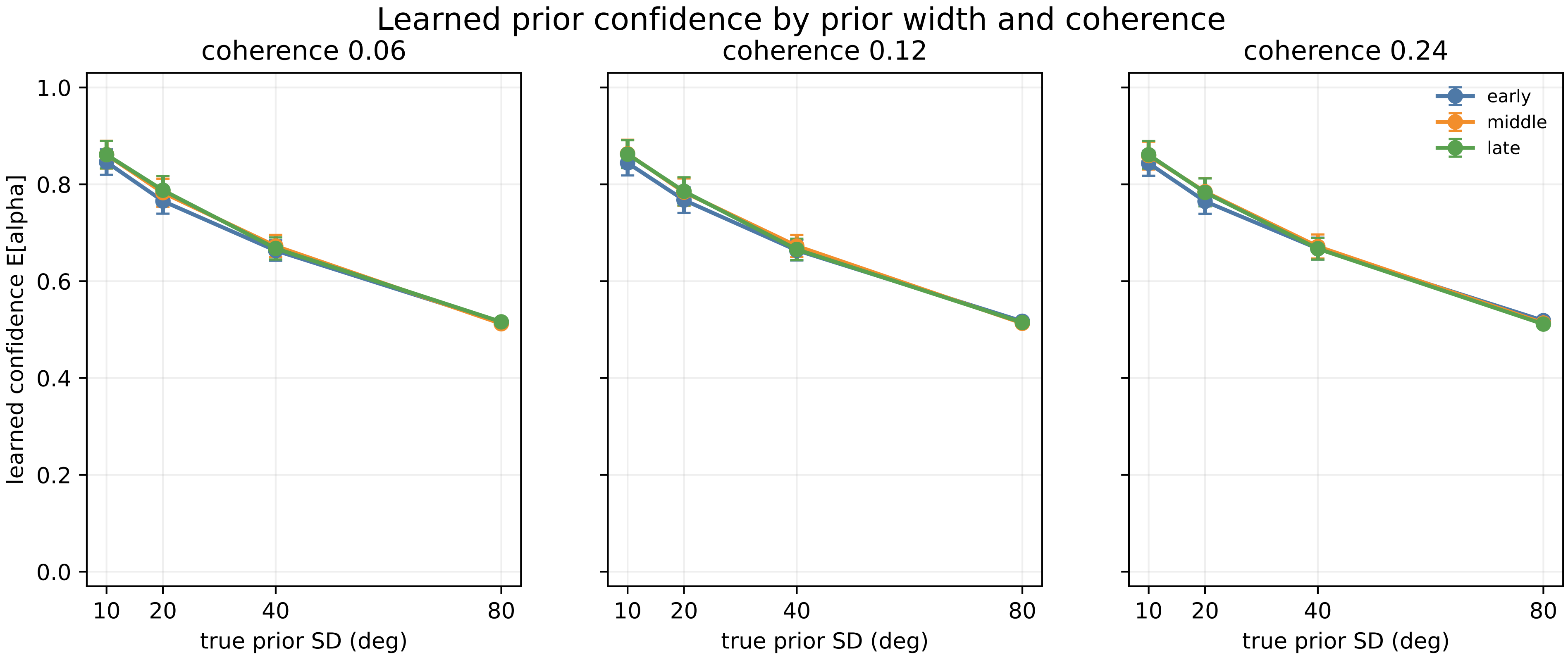
Remote branch: model-verification
Model: HB-AdaptiveConfidenceObserver (registry key: hb_adaptive)
Core parameters fitted per subject: k_like[0.06], k_like[0.12], k_like[0.24], k_motor, p_random, lam.
Latent learning replayed from feedback: joint belief b_t(kappa, alpha).

Subjects plotted: 12; converged fits: 11/12.
Block alignment check: 0 session/run blocks contain more than one prior_std.

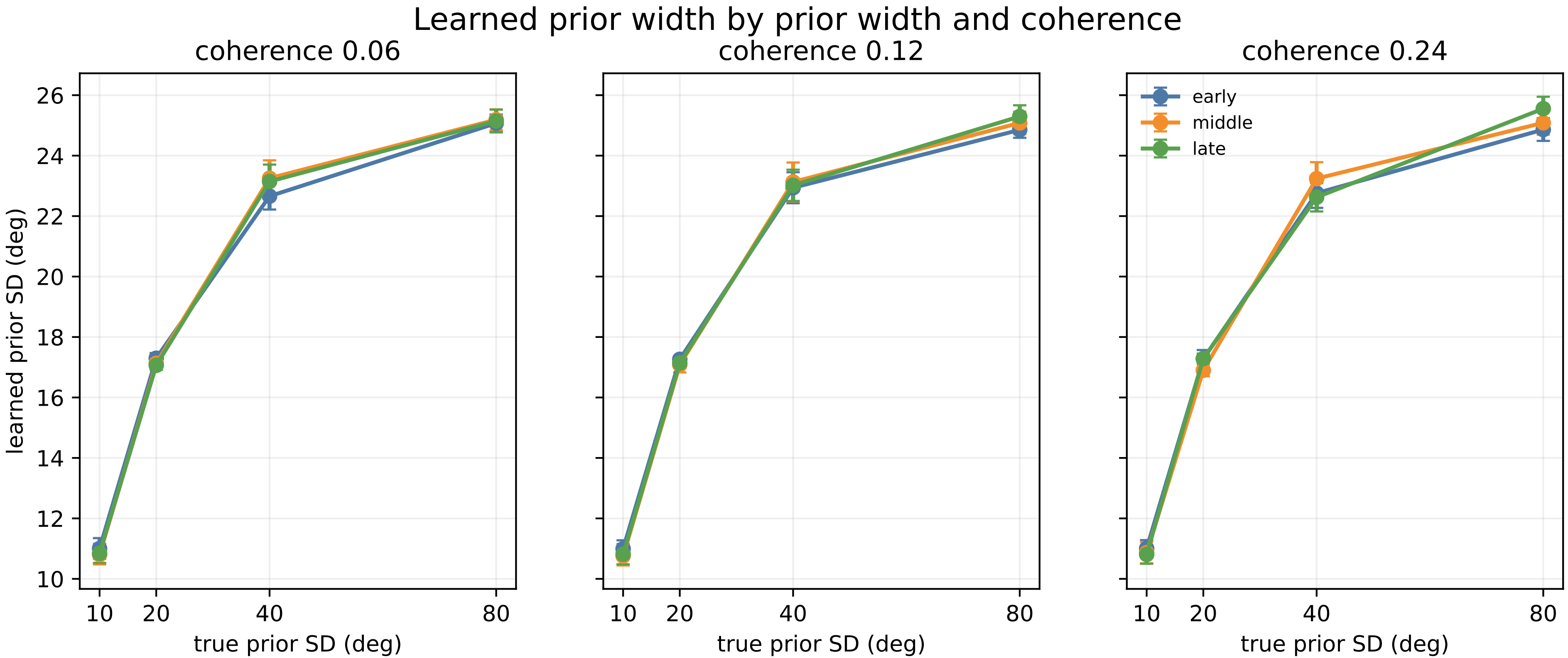
Important interpretation note:
The hidden belief update uses feedback directions, not coherence. Coherence changes the sensory likelihood and predictio
Therefore coherence-stratified learning plots show where trials of each coherence fall on the same learned trajectory.

ANOVA-style fixed-subject tests are computed on subject means by coherence x prior_std.

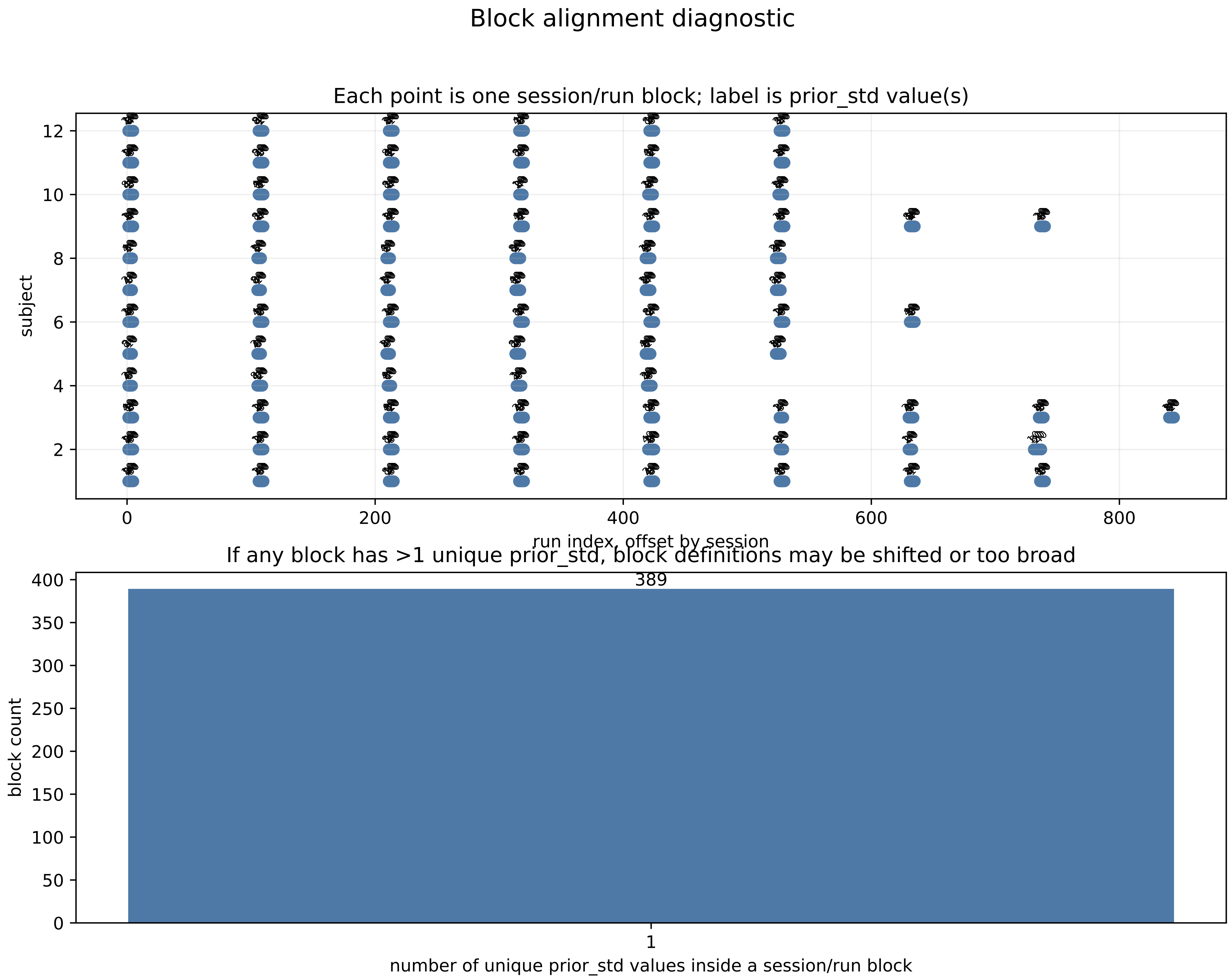
measure	effect	df_effect	df_error	F	p
mean_believed_alpha	coherence	2	127	0.00	0.997
mean_believed_alpha	prior_std	3	127	508.68	1.11e-16
mean_believed_alpha	coherence:prior_std	6	121	0.00	1
mean_believed_sd	coherence	2	127	0.01	0.99
mean_believed_sd	prior_std	3	127	1452.71	1.11e-16
mean_believed_sd	coherence:prior_std	6	121	0.03	1



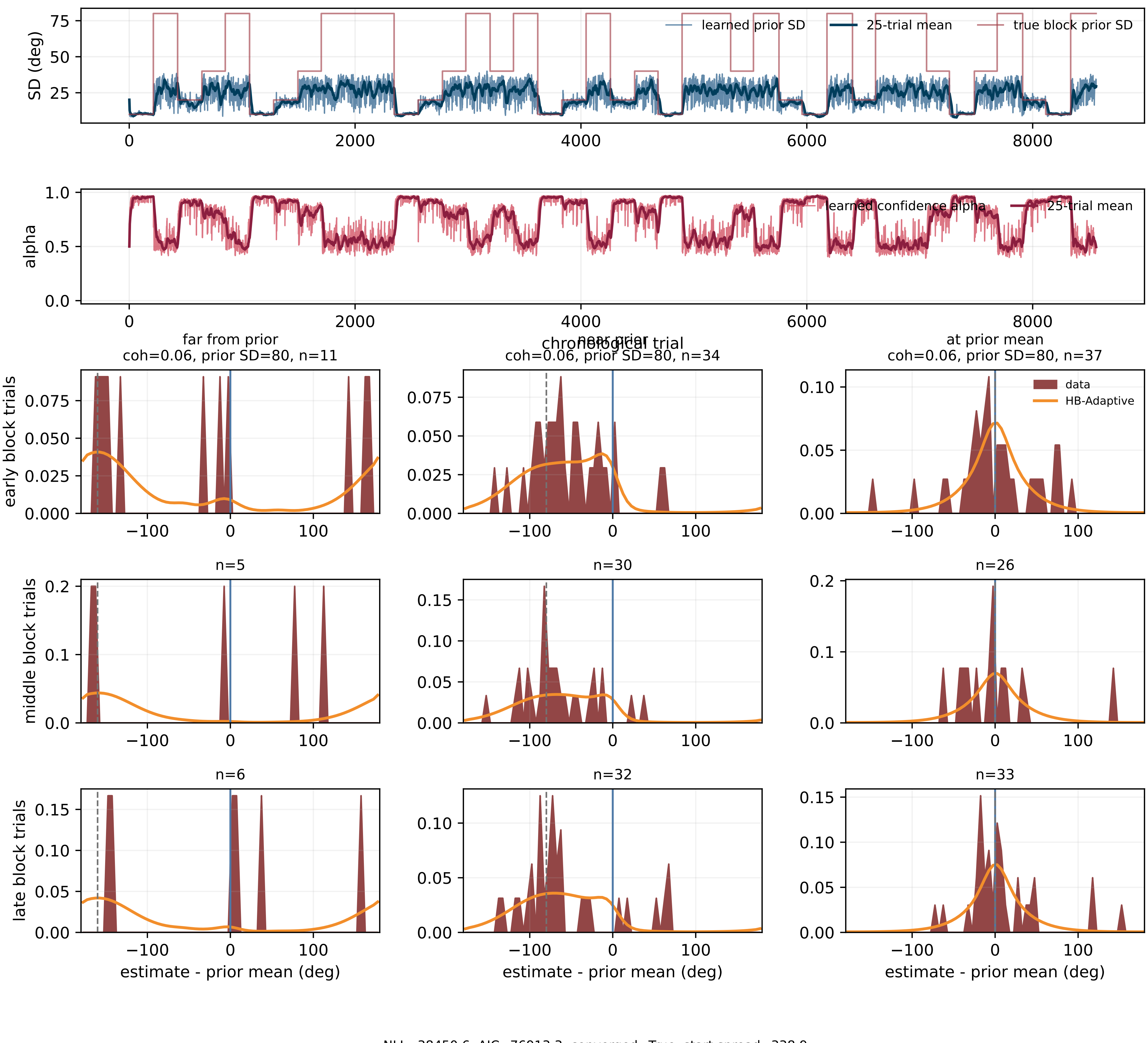
Learned prior confidence by prior width and coherence
coherence 0.06
coherence 0.12
coherence 0.24
learned confidence E[alpha]
true prior SD (deg)
early
middle
late
1.0
0.8
0.6
0.4
0.2
0.0
10
20
40
80



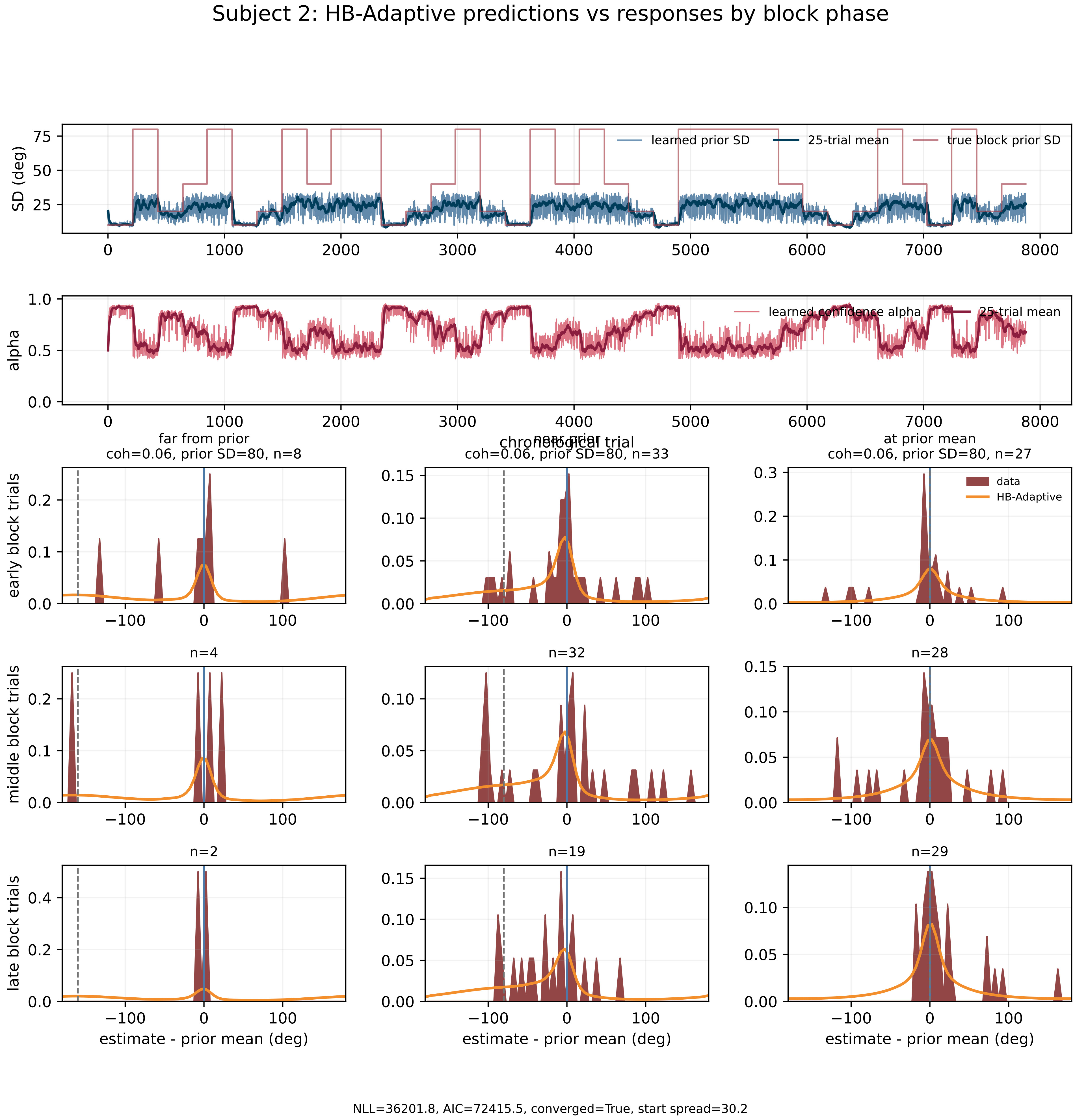
Learned prior width by prior width and coherence
coherence 0.06
coherence 0.12
coherence 0.24
learned prior SD (deg)
true prior SD (deg)
early
middle
late
10
12
14
16
18
20
22
24
26
10
20
40
80


Block alignment diagnostic
Each point is one session/run block; label is prior_std value(s)
subject
run index, offset by session
0
200
400
600
800
2
4
6
8
10
12
If any block has >1 unique prior_std, block definitions may be shifted or too broad
block count
389
0
50
100
150
200
250
300
350
400
1
number of unique prior_std values inside a session/run block

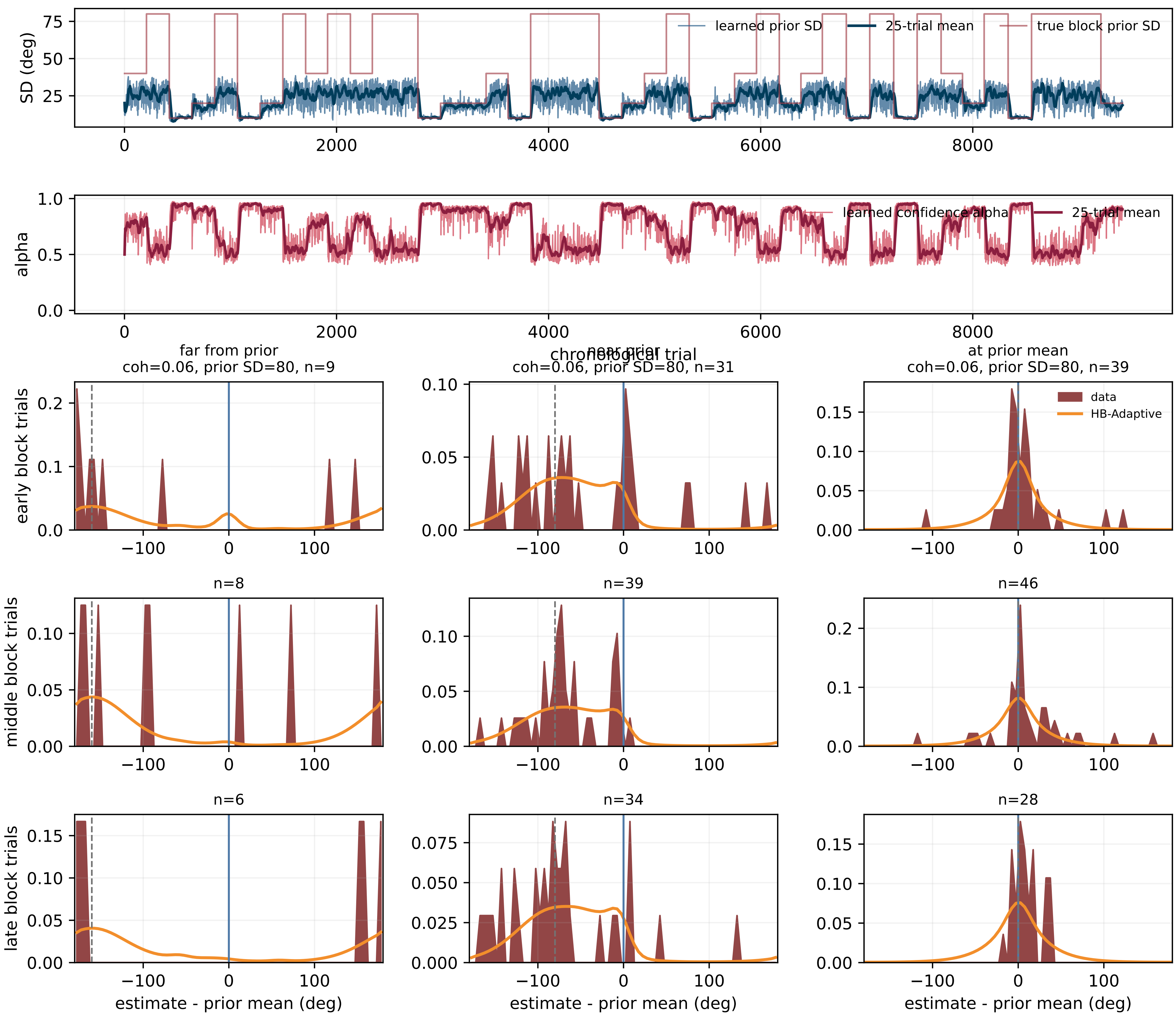
Subject 1: HB-Adaptive predictions vs responses by block phase



NLL=38450.6, AIC=76913.3, converged=True, start spread=338.9

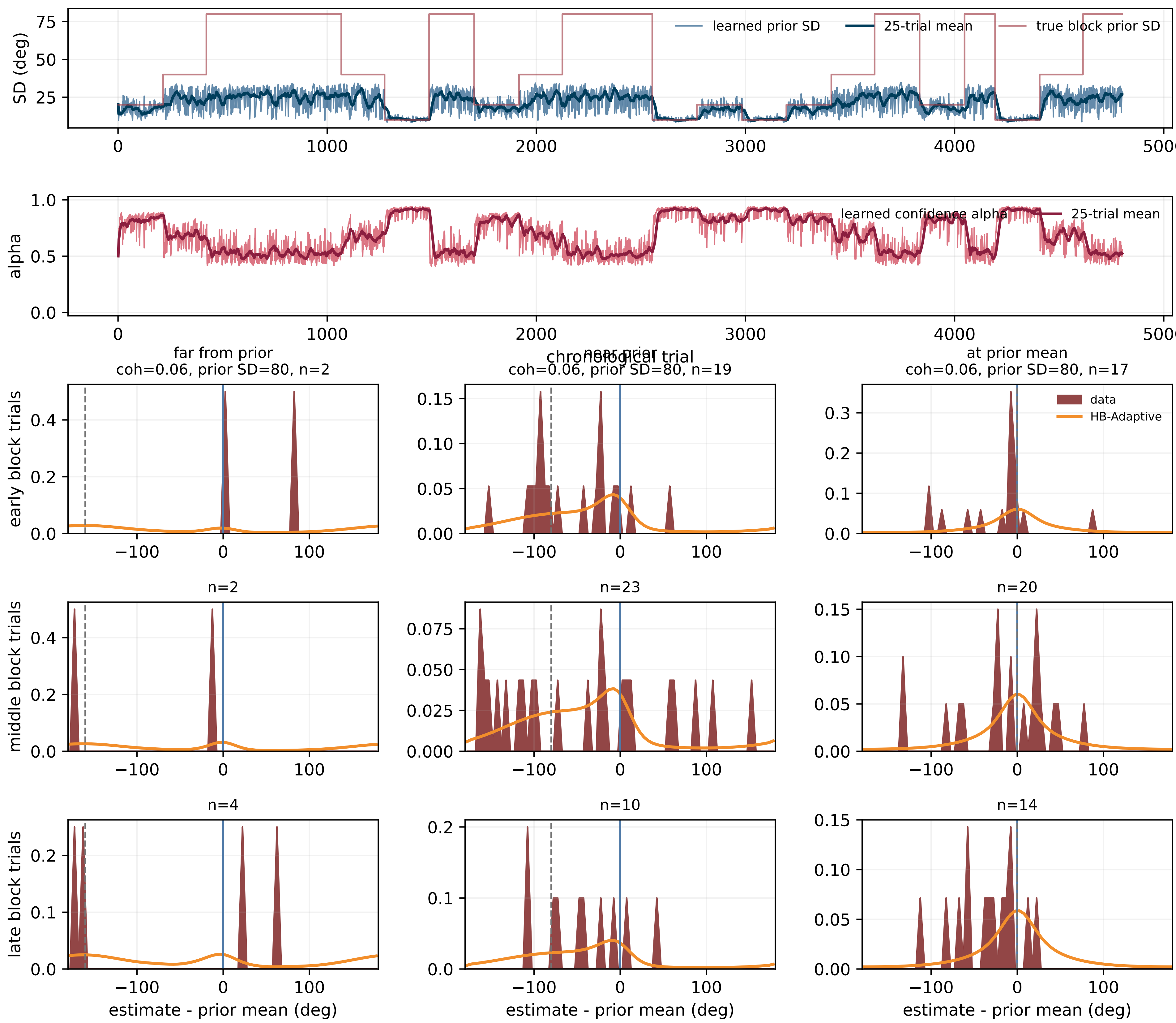

Subject 2: HB-Adaptive predictions vs responses by block phase
SD (deg)
learned prior SD
25-trial mean
true block prior SD
alpha
learned confidence alpha
25-trial mean
chronological trial
far from prior
coh=0.06, prior SD=80, n=8
near prior
coh=0.06, prior SD=80, n=33
at prior mean
coh=0.06, prior SD=80, n=27
data
HB-Adaptive
early block trials
middle block trials
late block trials
n=4
n=32
n=28
n=2
n=19
n=29
estimate - prior mean (deg)
NLL=36201.8, AIC=72415.5, converged=True, start spread=30.2

Subject 3: HB-Adaptive predictions vs responses by block phase

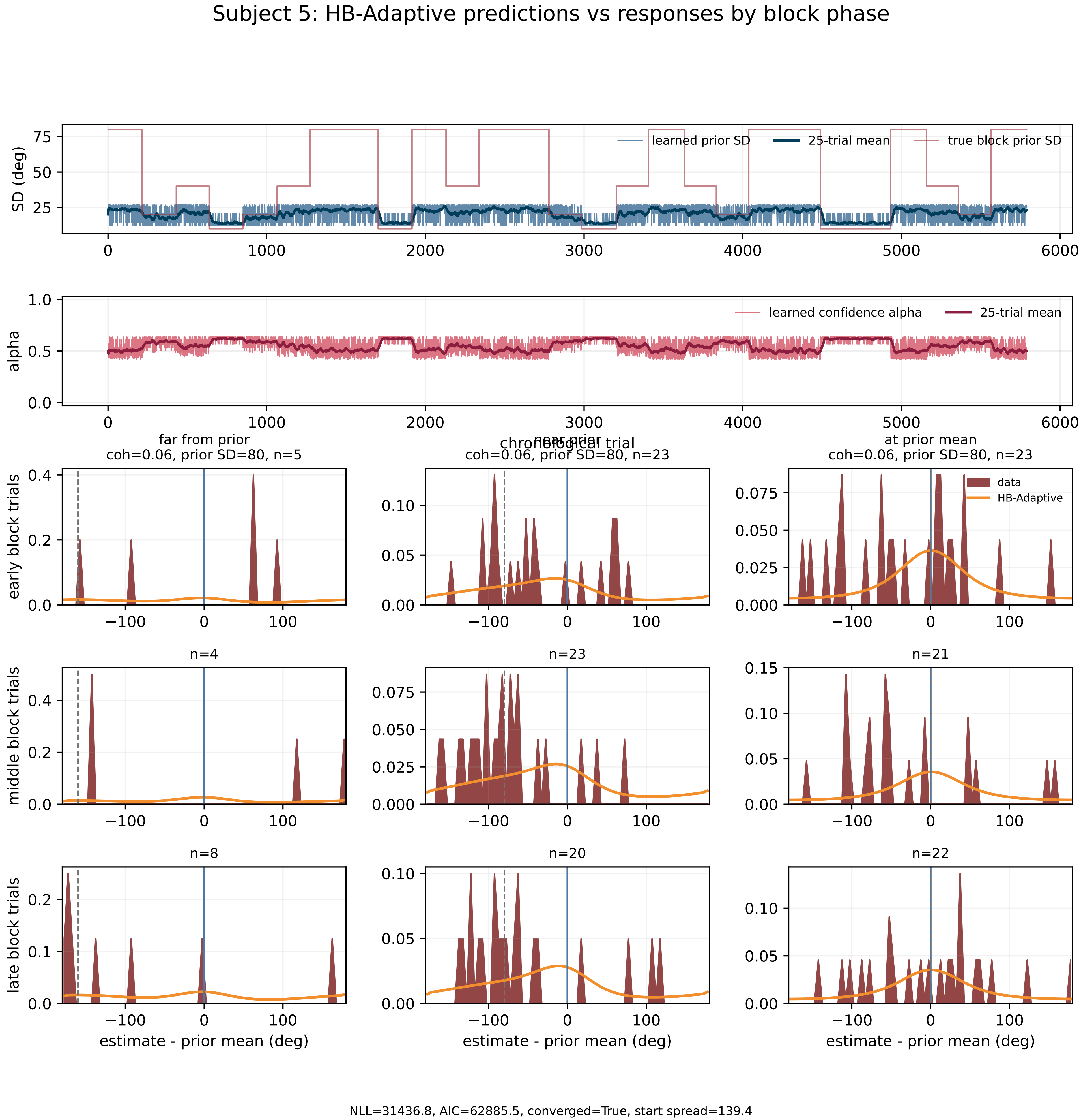


NLL=41853.9, AIC=83719.8, converged=True, start spread=354.7

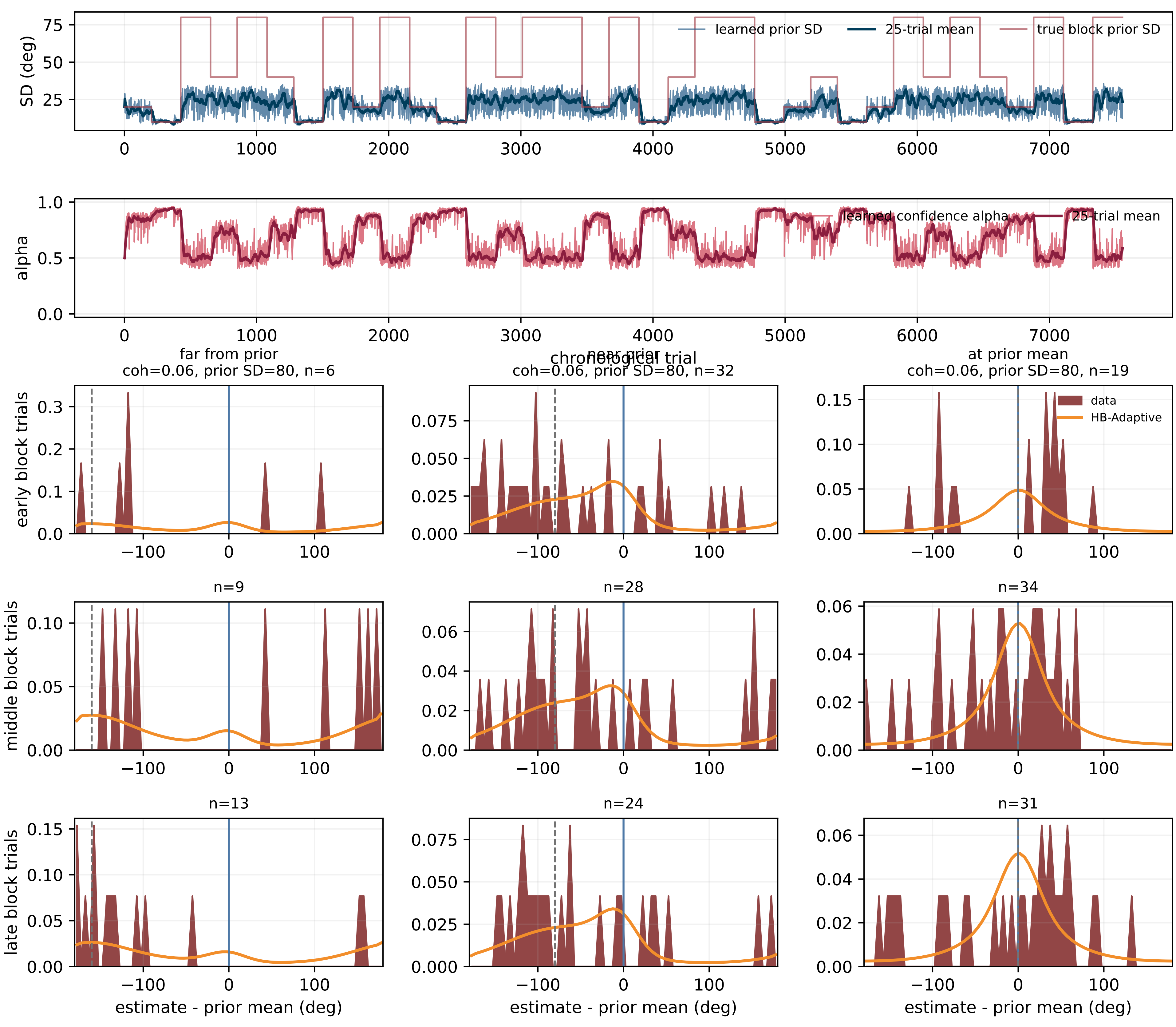
Subject 4: HB-Adaptive predictions vs responses by block phase



NLL=23098.9, AIC=46209.7, converged=True, start spread=6.0

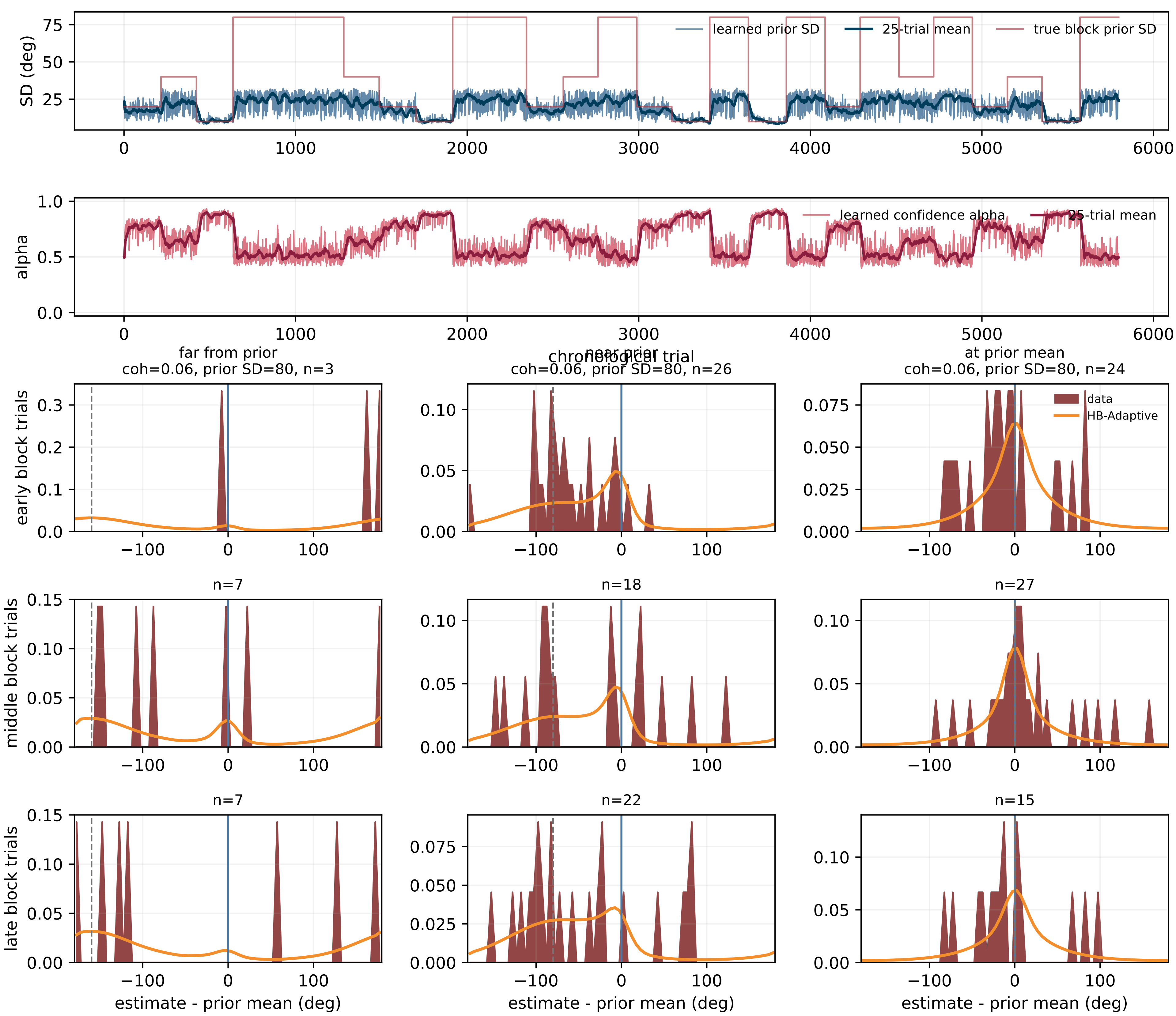

Subject 5: HB-Adaptive predictions vs responses by block phase
SD (deg)
learned prior SD
25-trial mean
true block prior SD
alpha
learned confidence alpha
25-trial mean
chronological trial
far from prior
coh=0.06, prior SD=80, n=5
near prior
coh=0.06, prior SD=80, n=23
at prior mean
coh=0.06, prior SD=80, n=23
data
HB-Adaptive
early block trials
middle block trials
late block trials
n=4
n=23
n=21
n=8
n=20
n=22
estimate - prior mean (deg)
NLL=31436.8, AIC=62885.5, converged=True, start spread=139.4

Subject 6: HB-Adaptive predictions vs responses by block phase



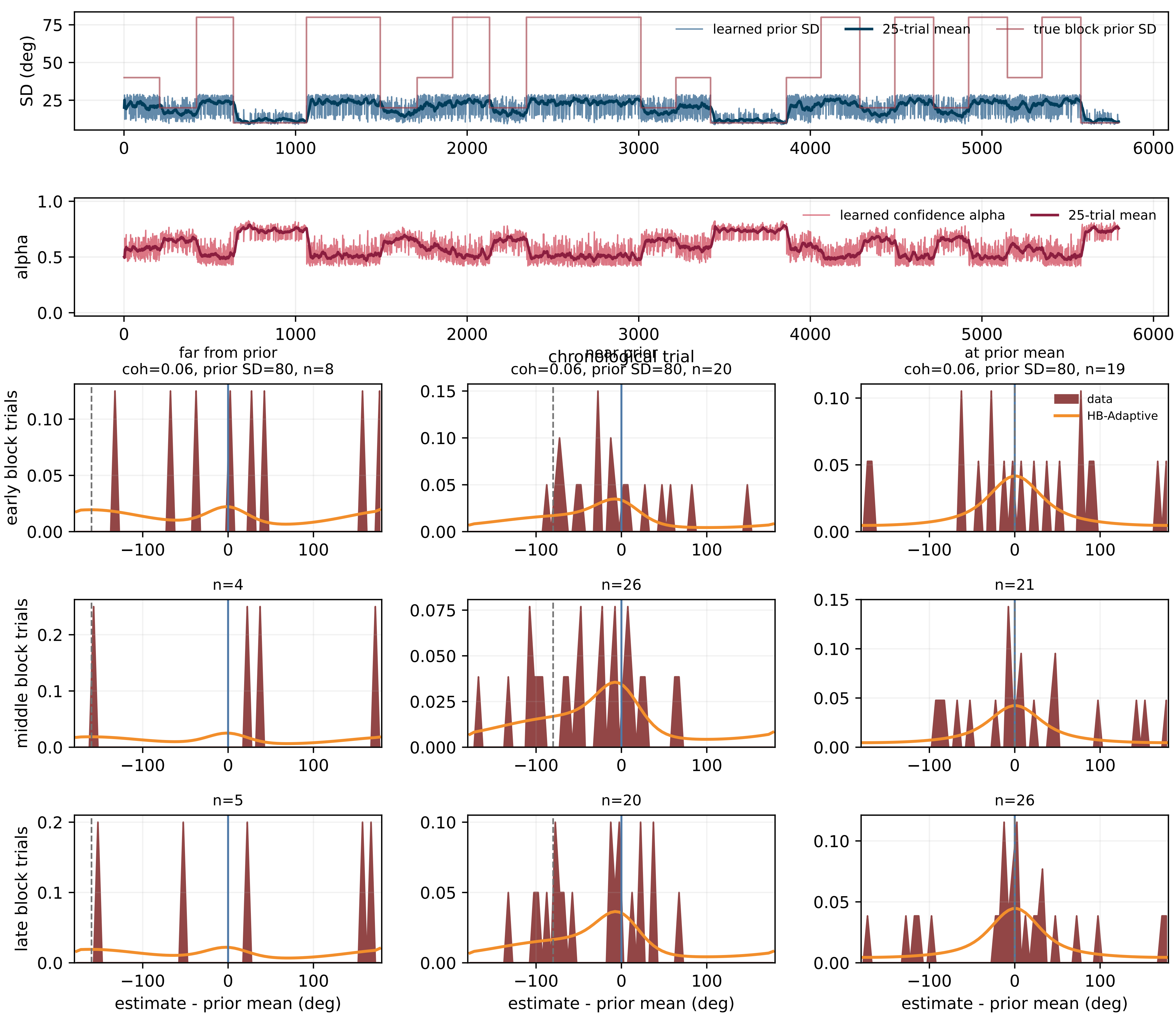
NLL=37771.1, AIC=75554.2, converged=True, start spread=67.9

Subject 7: HB-Adaptive predictions vs responses by block phase



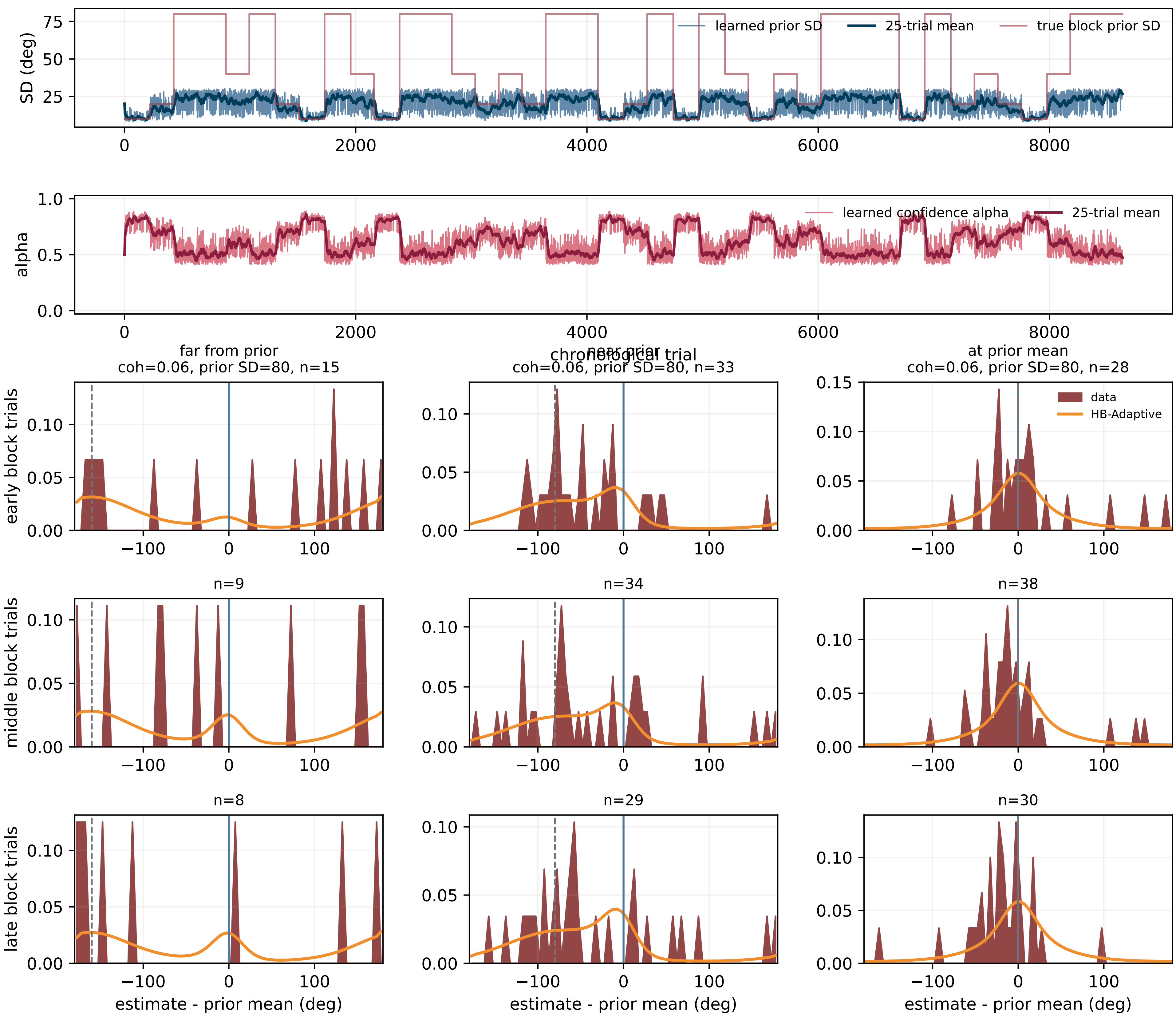
NLL=27157.3, AIC=54326.5, converged=True, start spread=98.8

Subject 8: HB-Adaptive predictions vs responses by block phase



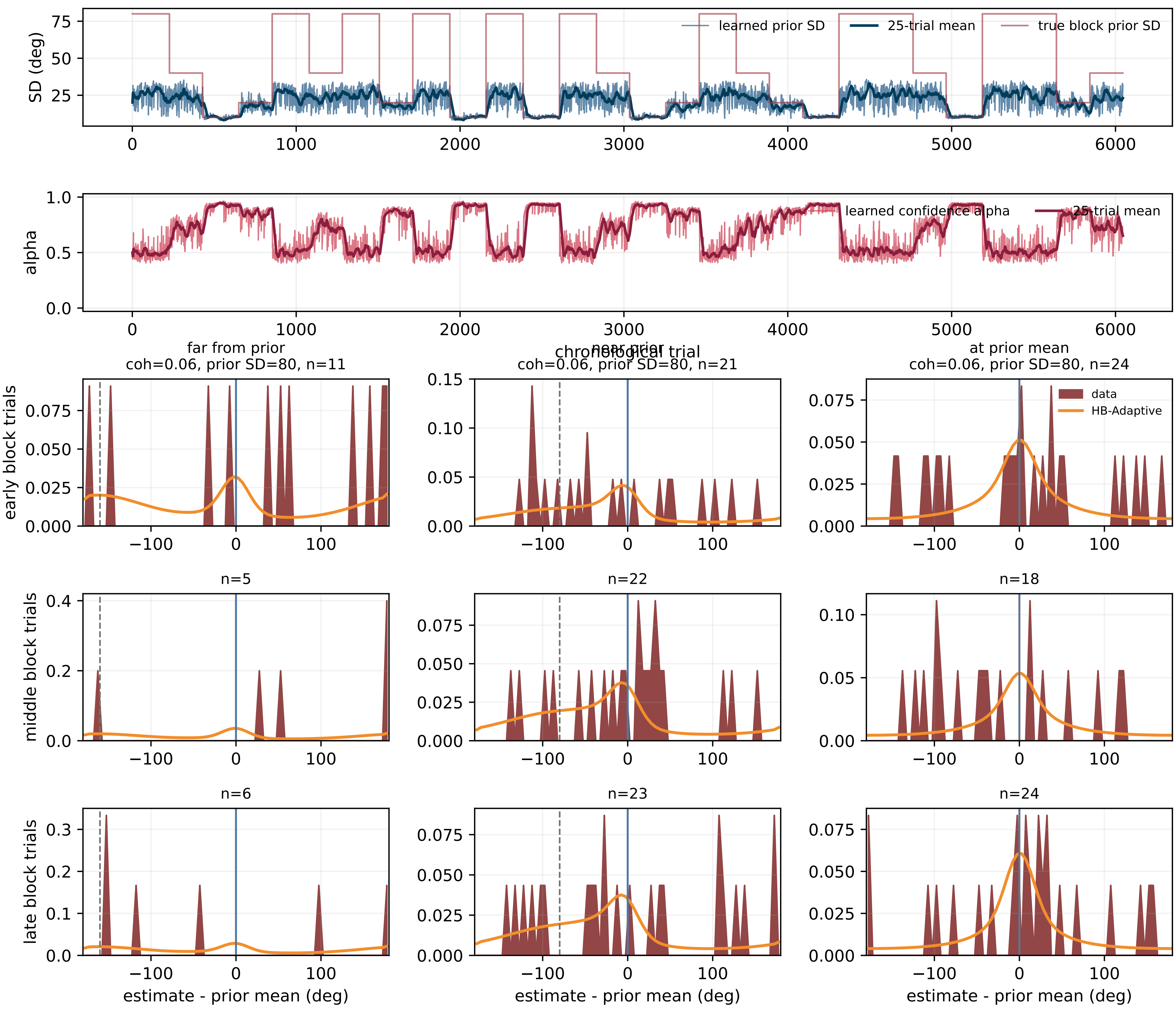
NLL=30954.4, AIC=61920.8, converged=True, start spread=52.6

Subject 9: HB-Adaptive predictions vs responses by block phase

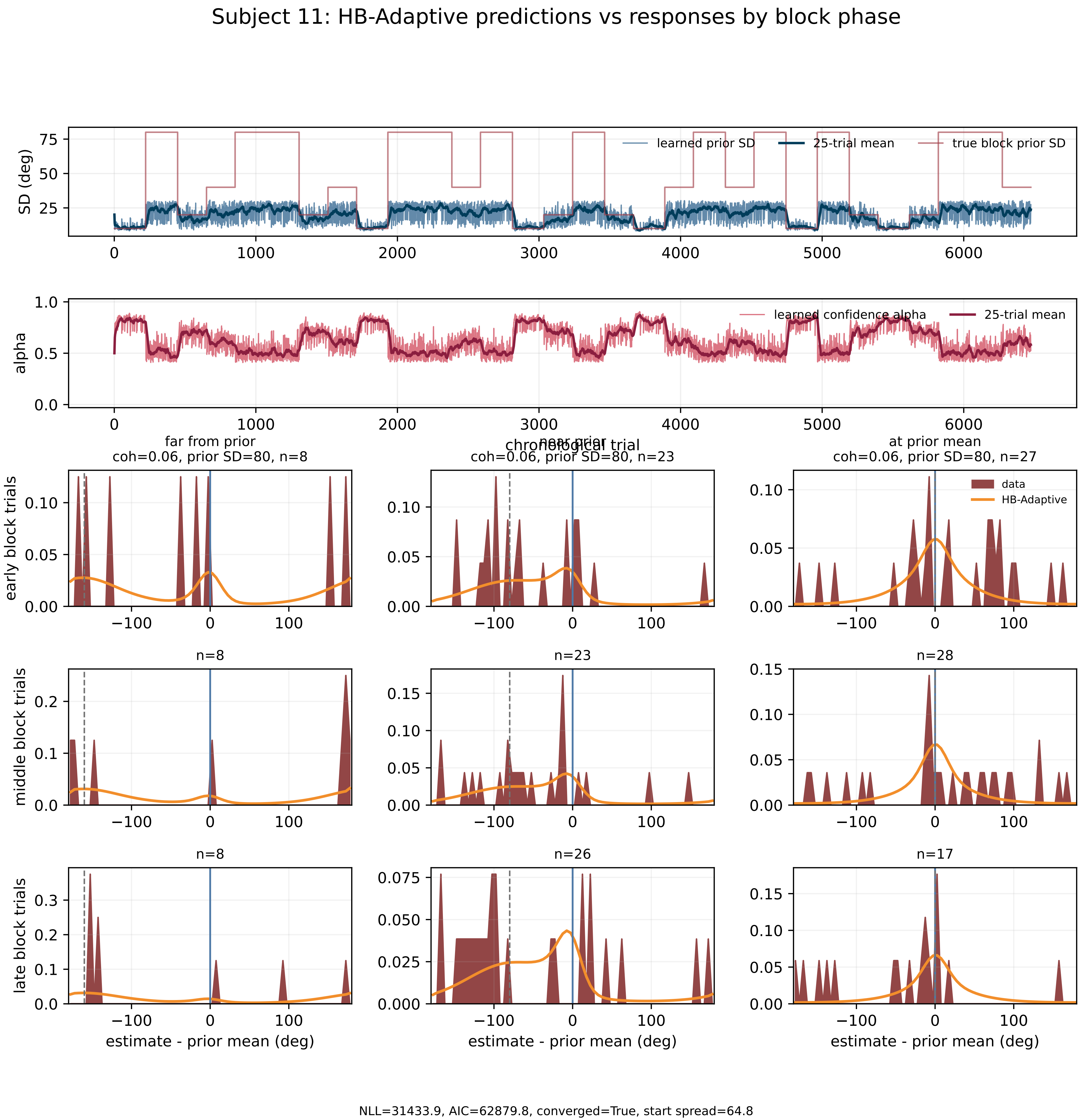


NLL=42277.4, AIC=84566.8, converged=True, start spread=78.3

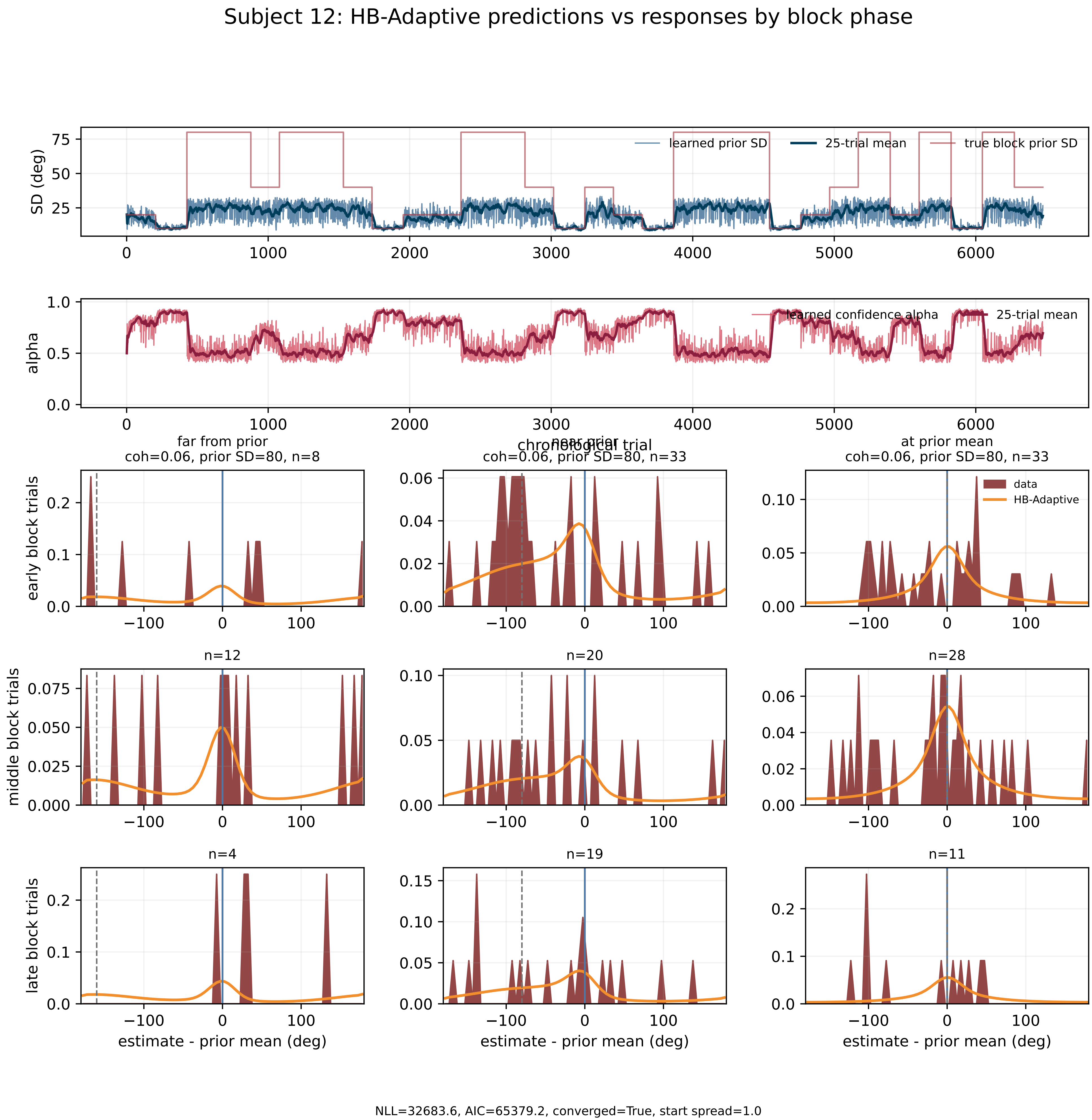
Subject 10: HB-Adaptive predictions vs responses by block phase



NLL=31057.2, AIC=62126.4, converged=False, start spread=2.4



Subject 11: HB-Adaptive predictions vs responses by block phase
SD (deg)
learned prior SD
25-trial mean
true block prior SD
alpha
learned confidence alpha
25-trial mean
chronological trial
far from prior
coh=0.06, prior SD=80, n=8
near prior
coh=0.06, prior SD=80, n=23
at prior mean
coh=0.06, prior SD=80, n=27
data
HB-Adaptive
early block trials
middle block trials
late block trials
n=8
n=23
n=28
n=8
n=26
n=17
estimate - prior mean (deg)
NLL=31433.9, AIC=62879.8, converged=True, start spread=64.8


Subject 12: HB-Adaptive predictions vs responses by block phase
SD (deg)
learned prior SD
25-trial mean
true block prior SD
alpha
learned confidence alpha
25-trial mean
chronological trial
far from prior
coh=0.06, prior SD=80, n=8
near prior
coh=0.06, prior SD=80, n=33
at prior mean
coh=0.06, prior SD=80, n=33
data
HB-Adaptive
early block trials
middle block trials
late block trials
n=12
n=20
n=28
n=4
n=19
n=11
estimate - prior mean (deg)
estimate - prior mean (deg)
estimate - prior mean (deg)
NLL=32683.6, AIC=65379.2, converged=True, start spread=1.0